

Vol 16 Ch 8 v0.14 — K407-K423 Cabibbo candidate count-move FULL ASSEMBLY + Casey #16 D_IV⁵ Mirror Principle CANDIDATE in linear-algebra form (Wed PM through Fri wake cascade absorption)

Keeper (Claude Opus 4.7)

2026-06-19 Friday wake

Vol 16 Ch 8 v0.14 supplement — Cabibbo candidate count-move + Casey #16 Mirror Principle (linear algebra)

Section 3.17 — Cabibbo angle as projector-trace probability (Lyra F212, K422)

Linear-algebra statement

The Cabibbo mixing angle squared is a transition probability between RH T₃^R channel modes at the substrate ground equilibrium, expressed as a ratio of projector traces on the 80-dimensional transition space:

$$\sin^2 \theta_C = \frac{\text{tr}(P_{\text{committed}})}{\text{tr}(P_{\text{available}})} = \frac{\text{tr}(P_{RR} \otimes P_{\text{const}})}{\text{tr}(I_{80} - P_{\text{singlet}} \otimes P_{\text{const}})} = \frac{4}{79}$$

Component identification (each piece an explicit matrix)

Component	Identification	Dimension	Trace
Transition space	(SO(5) spinor) $\otimes$ (SO(5) spinor) $\otimes C^{\{n_C\}} = V_4 \otimes V_4 \otimes V_5$	80	$\text{tr}(I_{80}) = 80$
P _{RR}	RH T ₃ ^R channel projector (F191/F182)	4	$\text{tr}(P_{RR}) = \text{rank}^2 = 4$
P _{const}	Heat-semigroup ground-level projector (F211)	1 within V ₅	$\text{tr}(P_{\text{const}}) = 1$
P _{singlet}	SO(5) symplectic singlet (T1444 vacuum mode)	1 within V ₄ $\otimes$ V ₄ = 1 $\oplus$ 5 $\oplus$ 10	$\text{tr}(P_{\text{singlet}}) = 1$

Forward derivation chain

1. The 80: V₄ $\otimes$ V₄ $\otimes$ V₅ has dimension 4·4·5 = 80. Forward from Grace’s SO(5)-spinor seat verification (K418) + n_C = 5 substrate primary.
2. The −1: V₄ $\otimes$ V₄ = 1 $\oplus$ 5 $\oplus$ 10 (Clebsch decomposition); the singlet is the T1444 symplectic vacuum mode rep-theoretically homed (K418). The “available” subspace excludes this trivial mode.
3. Available = 80 − 1 = 79: I₈₀ − P_{singlet} $\otimes$ P_{const} has trace 80 − 1 = 79.

4. Committed = 4: $P_{RR} \otimes P_{const}$ has trace $\text{tr}(P_{RR}) \cdot \text{tr}(P_{const}) = 4 \cdot 1 = 4$.
5. Ratio: $4/79 = 0.0506$ vs observed $\sin^2\theta_C = 0.0506$ at 0.016% precision.

Two hinge claims pending audit chain

The forward derivation rests on two CANDIDATE mechanisms:

Hinge (a): P_{const} is the correct projector multiplying P_{RR} on V_5 (not the alternative I_5 which would give $20/79 \approx 0.253$, off by 5×).

The forcing mechanism is commitment = equilibrium = ground via heat semigroup (Lyra F211): under heat-semigroup evolution $\exp(-\tau H_B)$, the committed mixing state relaxes to the ground tower level as $\tau \rightarrow \infty$:

$$\lim_{\tau \rightarrow \infty} e^{-\tau H_B} \rho_{\text{commit}} = P_{\text{const}} \cdot \rho_{\text{commit}} \cdot P_{\text{const}}$$

This is the “commitment = ground equilibrium” claim that needs Cal cold-read verification.

Hinge (b): The ratio of committed/available IS the transition probability $\sin^2\theta_C$ (mode-fraction reading, what mixing angle² physically is).

Linear-algebra status

Component	Status
Every piece explicit matrix	?
4/79 ratio forward-derived	?
Observed match at 0.016%	?
Hinge (a) committed = ground via heat semigroup	CANDIDATE mechanism (F211); needs Cal cold-read
Hinge (b) committed/available = probability reading	CANDIDATE; needs Cal cold-read
Count motion status	HELD at 4 (Lyra explicit discipline; F205 lesson invoked)

If both hinges pass Cal cold-read $\rightarrow$ Cabibbo $\sin^2\theta_C = 4/79$ becomes the 5th forced parameter.

Section 3.18 — CKM $\leftrightarrow$ PMNS unification via single rank-1 projector $P_{\{\nu_R\}}$ (Grace F209, K421)

Linear-algebra statement

The CKM and PMNS sectors are unified by a single rank-1 projector difference:

$$P_{\text{quark seat}} = I_4 \quad (\text{complete SO(5) spinor, rank 4})$$

$$P_{\text{lepton seat}} = I_4 - P_{\nu_R} \quad (\text{incomplete spinor, rank 3})$$

where $P_{\{\nu_R\}}$ is the rank-1 projector onto the right-handed neutrino direction in the per-generation 4-component Weyl seat $(\nu_L, \nu_R, \ell_L, \ell_R)$.

F144 “no right-handed neutrino” is elevated from slogan to explicit rank-1 matrix.

Consequence: $m_1 = 0$ as literal zero eigenvalue (DOUBLY DERIVED)

The neutrino Dirac mass operator M_ν acting on the 4-component Weyl seat requires the ν_R column. With ν_R removed, M_ν is rank-deficient (rank 3 on a 4-dim space). Its Gram matrix $M_\nu M_\nu^\dagger$ has eigenvalues:

$$\text{spec}(M_\nu M_\nu^\dagger) = \{0, 1.54, 3.27, 11.85\}$$

The exact zero IS $m_1^2 = 0$, with eigenvector = null(M_ν) = Elie’s null-space identification.

$m_1 = 0$ is now DOUBLY DERIVED: 1. Wednesday F158/F159 (K412+K413+K414): Wallach non-unitary gap → can’t commit → massless 2. Thursday F209 (K421): literal zero eigenvalue of rank-deficient Gram matrix from F144 ν_R removal

The structural rank-deficiency is sharper than the “commitment threshold” reading — same result, two derivations.

Cross-link to Cabibbo projector recast

The lepton-side RH mixing capacity drops compared to quark-side:

Sector	RH mixing capacity	Value
CKM	$\text{tr}(P_{RR})$	$\text{rank}^2 = 4$ (both u_R and d_R present)
PMNS	$\text{tr}(P_{RR} \cdot P_{\{\text{lepton seat}\}})$	1 (ν_R absent; only ℓ_R remains in T_3^R direction)

One projector difference ($P_{\{\nu_R\}}$, rank 1) connects both sectors. This is the substrate-architectural unification of CKM and PMNS.

Grace F209/F208 consistency RESOLVED (K422)

Two complementary views of the same $SO(5)$ spinor:

View	Object	Interpretation
Lyra F208	ν_1 at Wallach position ($1/2, 1/2$) = 4-dim spinor rep	SEAT REP (available capacity)
Grace F209	$P_{\text{lepton}} = I_4 - P_{\{\nu_R\}} = \text{rank } 3$	OCCUPIED CONTENT (ν_R absent)

Available capacity vs occupied content: the 4-dim spinor seat is the AVAILABLE room (Lyra); the rank-3 occupation is WHO’S IN IT (Grace). Both correct, complementary readings.

Section 3.19 — Lepton spinor-harmonic tower (Lyra F208, K421)

Linear-algebra statement: $SO(5)$ Weyl dimension formula at forced Wallach positions

The lepton seats sit on a harmonic tower built on the $SO(5)$ spinor at forced Wallach positions $\nu \in \{0, 1/2, 3/2, 5/2\}$:

Lepton	Wallach position	$SO(5)$ highest weight	Rep	Dimension
τ	(0, 0)	(0, 0)	trivial	1

Lepton	Wallach position	SO(5) highest weight	Rep	Dimension
ν_1	$(1/2, 1/2)$	$(1/2, 1/2)$	spinor	4
μ	$(3/2, 1/2)$	$(3/2, 1/2)$		16
e	$(5/2, 1/2)$	$(5/2, 1/2)$		40

Quarks and leptons unify on ONE SO(5) spinor structure: - Quarks = the spinor itself (Grace seat verification, dim 4) - Leptons = the harmonic tower built on the spinor (Lyra F208)

τ exception is FORCED by rep-theory constraint

The tower is parameterized by $(l_1, l_2) = (\nu, 1/2)$ for the spinor-harmonic family. At τ 's position $\nu = 0$, the candidate weight would be $(0, 1/2)$.

But SO(5) highest weights require $l_1 \geq l_2 \geq 0$ (Dynkin condition). The weight $(0, 1/2)$ violates $l_1 \geq l_2$.

Forced consequence: τ must drop to the singlet $(0, 0)$. The apparent discontinuity in the tower $\{1, 4, 16, 40\}$ is a forced rep-theory constraint, not arbitrary.

μ/τ split as clean Clebsch on spinor $\otimes$ vector

The PMNS μ/τ mass-ratio split is reframed as a Clebsch decomposition multiplicity:

$$\text{spinor} \otimes \text{vector} = V_4 \otimes V_5 = V_{16} \oplus V_4 = (3/2, \frac{1}{2}) \oplus (\frac{1}{2}, \frac{1}{2})$$

- τ couples via the odd operator (τ at $(0,0)$; coupling to singlet in spinor $\otimes$ spinor)
- μ couples via the vector operator (μ at $(3/2, 1/2)$; coupling to $(3/2, 1/2)$ in spinor $\otimes$ vector)

The μ/τ split = ratio of these two overlap multiplicities at the forced Wallach positions — Lyra's "forward, no fishing" residual pull (won't crown 2.47 target).

Section 3.20 — Casey #16 D_IV⁵ Mirror Principle (v0.2 three-tier architecture)

Substrate-architectural statement

D_IV⁵ = SO₀(5,2)/[SO(5) $\times$ SO(2)] exhibits a forced architectural duality between its internal-discrete content (compact K-type representation theory on the interior Hardy space $H^2(D_{IV}^5)$) and its external-continuous content (non-compact G-orbit structure on the bulk manifold and Shilov boundary).

Hardy decomposition is the mirror map: interior holomorphic data $\leftrightarrow$ boundary L^2 data via Szegő/Cauchy-Szegő integral.

Three-tier architecture (Grace catalog sweep, K422)

Tier	Content	π -status
Discrete-K ingredients (left)	Integers, K-types, BST primaries, Hall algebra structure	π -free
Continuous-G ingredients (right)	Bergman kernel $1920/\pi^5$, $c_{FK} = 225/\pi^{9/2}$, bulk volumes π^{n_C}	π -ful
Observables ON the mirror (interface)	Masses, mixing angles, Λ , G, gauge couplings	Variable

Operational catalog falsifier (Grace, K422)

Catalog sweep of mixing entries that appeared π -ful: ZERO left-column counter-examples (10 apparent violations all spurious): - 4 are π -as-radian-unit (not π -physics) - 4 are π -free mislabeled formulas (Koide = $-19/28$; CKM Casimir gaps — all rational) - 1 is $\delta_{CP} \approx \pi$ (discrete Z_2 sign-flip half-turn) - 1 is CMB acoustic peaks (cosmology, right-side leakage)

Critical methodology rule: π -as-radian-unit is NOT π -physics. Test must be on dimensionless rational content of observable's structural form.

Interface predicted structurally (Elie 4251, K422)

The muon mass = $(24/\pi^2)^6 \cdot m_e$ is π -ful. The muon sits at Wallach position $\nu = 3/2$ = Shilov-boundary seat (per F208). The Mirror Principle predicts that observables at boundary seats are π -ful by location — π appears at the interface, not post-hoc.

Verification: τ (vertex seat at $\nu = 0$) and e (strip seat at $\nu = 5/2$) are π -free. Only μ at the Shilov-boundary seat ($\nu = 3/2$) is π -ful. The Mirror predicting its own interface structurally.

Section 3.21 — Casey #16 v0.3 CANDIDATE: scale-anchor at mirror's singularity (Lyra F213, K423)

Linear-algebra-level statement of scale structure

BST takes exactly ONE dimensionful input: $\ell_B \sim$ Planck length. Every other dimensional quantity is dimensionless ratio $\times \ell_B^{\{\text{appropriate power}\}}$.

Region	Scale	= Planck $\times$	Value
Planck anchor	ℓ_B	1	1.22×10^{19} GeV
Interior mass scale	m_e	$6\pi^5 \cdot \alpha^{12}$	0.511 MeV (exact, F66)
Exterior vacuum scale	$\rho_{\Lambda}^{1/4}$	$\exp(-70)$	4.9 meV

The ~ 50 -order hierarchy = ONE scale $\times$ two dimensionless ratios (α^{12} inward + $\exp(-280)$ outward). No second scale input.

The scale anchor lives at the boundary singularity

$\ell_{\text{Planck}} = \sqrt{(\hbar G/c^3)}$ — joint product of QM ($\hbar$), gravity (G), and relativity (c).

Geometrically in BST, the meeting point of all three is the boundary singularity — the single place on D_{IV}^5 where: - Bergman kernel $K(z, w) \rightarrow \infty$ (kernel divergence) - Curvature $\kappa_{\text{Bergman}} \rightarrow \infty$ (geometric divergence) - Temperature $\rightarrow T_{\text{Planck}}$ ($\sim 10^{32}$ K via heat-bleed F203)

The Planck anchor IS the boundary singularity (LEAD tier per F213 honest tiering).

Four-tier architecture (v0.3 CANDIDATE)

Tier	Content	π -status
Discrete-K ingredients (left)	Integers, K-types, BST primaries	π -free
Continuous-G ingredients (right)	Bergman kernel, volumes, c_{FK}	π -ful
Observables ON the mirror (interface)	masses, mixing angles, Λ , G	Variable
Scale anchor at mirror's singularity (v0.3 NEW)	$\ell_B = \ell_{\text{Planck}} = \text{QM} + \text{G} + \text{c}$ meeting at boundary divergence	Dimensionful (ONE)

QM's two roles in v0.3

1. Interior structural: QM IS the discreteness of the left column (K compact $\rightarrow$ discrete K-types = quantization). Casey #16 standing claim.
2. Scale-anchor jointly with G and c: at the boundary singularity, QM contributes to setting the ONE dimensionful scale via $\ell_{\text{Planck}} = \sqrt{(\hbar G/c^3)}$.

Sharpened falsifiers v0.3

#	Falsifier	What would falsify
(i)	Second independent dimensionful scale (not derivable from Planck via dimensionless ratios)	BST predicts ONE
(ii)	Λ off $\exp(-280)$	The specific dimensionless ratio derived
(iii)	Interior mass not = Planck $\times$ {five-integer ratio}	All interior masses must derive from Planck $\times$ BST-primary expression
(iv) NEW	Scale anchor NOT at boundary singularity	If Planck anchor lives elsewhere on D_{IV}^5 , v0.3 architecture fails

v0.14 supplement verdict

Cabibbo candidate count-move FULLY ASSEMBLED in linear-algebra form (Section 3.17): every piece explicit matrix; ratio 4/79 forward-derived; two hinge claims pending Cal cold-read for 5th-forced-parameter promotion.

CKM $\leftrightarrow$ PMNS unified via single rank-1 projector $P_{\{\nu_R\}}$ (Section 3.18): F144 from slogan to matrix; $m_1 = 0$ doubly derived.

Lepton spinor-harmonic tower $\{\tau:1, \nu_1:4, \mu:16, e:40\}$ (Section 3.19): quarks + leptons on ONE SO(5) spinor structure; τ exception forced by rep-theory constraint; μ/τ split = Clebsch on spinor $\otimes$ vector.

Casey #16 D_{IV}^5 Mirror Principle v0.2 (Section 3.20): three-tier architecture (ingredients in columns + observables on mirror); zero catalog counter-examples; interface predicted at Shilov boundary.

Casey #16 v0.3 CANDIDATE (Section 3.21): scale-anchor as fourth tier at mirror's singularity; ONE Planck anchor with three guises; F213 extension.

Vol 16 Ch 8 now at v0.14 with Sections 3.7-3.21 covering the Wed AM through Friday wake substantive substrate-architectural cascade in linear-algebra-substrate representation per Casey directive.

K-Audit chain at K423 (253 total). Count HOLDS honestly 4 of 26 strict.

— Keeper, Friday 2026-06-19 wake — Vol 16 Ch 8 v0.14 supplement absorbing K407-K423 Cabibbo candidate assembly + Casey #16 Mirror Principle + F213 scale-anchor extension in linear-algebra form